Started on	Monday, 4 November 2024, 3:18 PM
State	Finished
Completed on	Friday, 8 November 2024, 6:47 PM
Time taken	4 days 3 hours
Marks	30.00/30.00
Grade	**100.00** out of 100.00
Name	DHIRU BHAI 730423104017

Question 1

Correct

Mark 10.00 out of 10.00

Write a java program to perform the following operations.

- Create a Priority Queue PQ of type String
- Read the size of the queue from the user.
- Read the elements of the Queue
- Display the Queue PQ
- Read a string from the user and remove it from Queue and display PQ.

For example:

Input	Result
6 Swift Alto Baleno Brezza Creta Triber Brezza	Initial Queue [Alto, Brezza, Baleno, Swift, Creta, Triber] After Remove [Alto, Creta, Baleno, Swift, Triber]

Answer: (penalty regime: 0 %)

```
import java.util.*;
public class Main
{
    public static void main(String[] args)
    {
        Scanner s=new Scanner(System.in);
        PriorityQueue<String> PQ=new PriorityQueue<>();
        int a=s.nextInt();
        s.nextLine();
        for(int i=0;i<a;i++)
        {
            PQ.add(s.next());
        }
        String r=s.next();
        System.out.println("Initial Queue "+PQ);
        PQ.remove(r);
        System.out.println("After Remove "+PQ);
    }
}
```

	Input	Expected	Got	
✔	6 Swift Alto Baleno Brezza Creta Triber Brezza	Initial Queue [Alto, Brezza, Baleno, Swift, Creta, Triber] After Remove [Alto, Creta, Baleno, Swift, Triber]	Initial Queue [Alto, Brezza, Baleno, Swift, Creta, Triber] After Remove [Alto, Creta, Baleno, Swift, Triber]	✔

Passed all tests! ✔

▸ **Show/hide question author's solution (Java)**

Correct

Marks for this submission: 10.00/10.00.

Question 2

Correct

Mark 10.00 out of 10.00

Create a stack S1 of type Integer and perform the following operations on it.

1. Read elements and add it to stack until you receive -1 (-1 denotes end of input and should not be pushed to stack)
2. Remove the last element that was added to the stack
3. Print the current element that is at the top of the stack
4. Determine the number of elements currently present in the stack

You have to use Java collections Stack to implement it. You will not get marks if you use any other java collections framework

For example:

Input	Result
5 7 9 4 6 3 1 2 -1	After removing Top Most value 1 3 6 4 9 7 5 Top of stack element: 1 The size of stack is 7

Answer: (penalty regime: 0 %)

```
import java.util.Scanner;
import java.util.Stack;
public class Main
{
    public static void main(String[] args)
    {
        Stack<Integer> S1 = new Stack<>();
        Scanner sc = new Scanner(System.in);
        while (true)
        {
            int num = sc.nextInt();
            if (num == -1) break;
            S1.push(num);
        }
        if (!S1.isEmpty())
        {
            S1.pop();
        }
        System.out.print("After removing Top Most value ");
        for (int i = S1.size() - 1; i >= 0; i--)
        {
            System.out.print(S1.get(i) + " ");
        }
        System.out.println();
        if (!S1.isEmpty())
        {
            System.out.println("Top of stack element: " + S1.peek());
        }
        System.out.println("The size of stack is " + S1.size());
    }
}
```

	Input	Expected	Got	
✔	5 7 9 4 6 3 1 2 -1	After removing Top Most value 1 3 6 4 9 7 5 Top of stack element: 1 The size of stack is 7	After removing Top Most value 1 3 6 4 9 7 5 Top of stack element: 1 The size of stack is 7	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Question 3

Correct

Mark 10.00 out of 10.00

Given the details of marks scored by students in form of a HashMap where the name of the student is the Key and the marks scored is the Value.

Your task is to sort the map according to the key values i.e the names of the students in the alphabetical(lexicographical) order.

Sample Input:

5

Akash 85

Vikram 95

Dev 76

Arun 92

Jai 81

Sample Output:

Akash 85

Arun 92

Dev 76

Jai 81

Vikram 95

For example:

Input	Result
5 Akash 85 Vikram 95 Dev 76 Arun 92 Jai 81	Akash 85 Arun 92 Dev 76 Jai 81 Vikram 95

Answer: (penalty regime: 0 %)

```
import java.util.*;

public class StudentMarksSorter {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        HashMap<String, Integer> marksMap = new HashMap<>();

        // Read the number of entries
        int n = scanner.nextInt();
        scanner.nextLine(); // Consume the newline character

        // Read student names and their marks
        for (int i = 0; i < n; i++) {
            String line = scanner.nextLine();
            String[] parts = line.split(" ");
            String name = parts[0];
            int marks = Integer.parseInt(parts[1]);
            marksMap.put(name, marks);
        }

```

```
        // Sort the entries based on the keys (student names)
        List<Map.Entry<String, Integer>> sortedEntries = new ArrayList<>(marksMap.entrySet());
        sortedEntries.sort(Map.Entry.comparingByKey());

        // Print the sorted output
        for (Map.Entry<String, Integer> entry : sortedEntries) {
            System.out.println(entry.getKey() + " " + entry.getValue());
        }

        scanner.close();
    }
}
```

	Input	Expected	Got	
✔	5 Akash 85 Vikram 95 Dev 76 Arun 92 Jai 81	Akash 85 Arun 92 Dev 76 Jai 81 Vikram 95	Akash 85 Arun 92 Dev 76 Jai 81 Vikram 95	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.